AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 17 · STUDENT LESSON

Initialisation and reproducibility

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why must an early-stopping report identify both the stop epoch and the restored checkpoint?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Trace how starting values affect a fixed training budget. Try the recall before opening notes.

Your learning target

I can trace how starting values affect a fixed **training** budget.

I can explain a simple hidden-unit **symmetry**.

I can record enough information to replay and audit a classroom experiment.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace how starting values affect a fixed **training** budget

Explain a simple hidden-unit **symmetry**

Record enough information to replay and audit a classroom experiment

Retrieve what you already know

> Why must an early-stopping report identify both the stop epoch and the restored checkpoint?

My starting explanation

This week’s end goal

By the end, I can explain **initialisation** and reproducibility using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Initialisation and reproducibility case. Separate observations, given facts and assumptions before choosing an explanation.

Return to h = ReLU(x), y = vh, **training** pairs (1, 2) and (2, 4), ordinary MSE and a frozen hidden unit. Only v is trained. Keep full-data updates, learning rate 0.1 and exactly two updates. Compare initial v values 0, 1 and 3.

This experiment changes only the start.

The gradient is 5(v - 2), so each step gives new v = 0.5v + 1. Signed distance from 2 halves on every update.

Different starting distances therefore produce different losses after this finite budget, even though all three runs approach the same minimum if this exact procedure continues.

Initial v	After one update	After two updates	Final training MSE
0	1	1.5	0.625
1	1.5	1.75	0.15625
3	2.5	2.25	0.15625

The equal losses for starts 1 and 3 arise because they are equally distant from 2. They do not have equal final parameters.

This simple convex example does not show the multiple solutions or inactive units that can occur in larger networks.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Initialisation and reproducibility. Connect each labelled input, deciding step and result.

A pseudorandom generator produces a sequence determined by its state. In the same Python setup, recreating a local generator with the same **seed** and making the same calls reproduces the sequence below.

Saving the generated values also removes ambiguity about which draw became which parameter.

```
from random import Random

def initial_values(seed):
    rng = Random(seed)
    return [rng.uniform(-1.0, 1.0) for _ in range(4)]

def train(v):
    states = [v]
    for _ in range(2):
        v = v - 0.1 * 5 * (v - 2)
        states.append(v)
    return states

first = initial_values(17)
second = initial_values(17)
assert first == second
print(first)
for start in [0.0, 1.0, 3.0]:
    print(start, train(start))
```

The four random numbers illustrate a saved **initialisation** sequence; they are not used in the three separate scalar **training** runs.

The range [-1, 1] is a classroom choice, not a general recommendation for network **initialisation**. An extra random draw before assigning weights shifts which value is used, even if the **seed** stays the same.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Initialisation and reproducibility. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Repeat the a = 0 run with identical arithmetic and settings.

> **Predicted result**
>
> **Prediction**: It again reaches 1 then 1.5 in this deterministic toy calculation.

Why the result follows

Reason: Identical initial state and update rules determine the same arithmetic trace.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Initialisation and reproducibility to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Initialisation	The procedure that assigns model parameter values before training
Seed	A value used to initialise a pseudorandom generator
Symmetry	An equivalence between parts of a model that can keep their computations identical
Reproducibility record	Saved data settings code and results that let another person check how an experiment was performed

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Initialisation and reproducibility, predict a step, then compare with its explanation.

Identical units receive identical updates

HIDDEN UNIT	BEFORE THE UPDATE	AFTER THE UPDATE
First ReLU unit	u is 1 and a is 1	u is 0.8 and a is 0.8
Second ReLU unit	v is 1 and b is 1	v is 0.8 and b is 0.8

In this separate two-unit example there are no biases. At x = 1, output is 2 and target is 1. Each of the four squared-error partial gradients equals 2. A simultaneous update at rate 0.1 changes every weight from 1 to 0.8 and keeps the units identical.

For x = 1, calculate the two-unit output and squared error before and after the shown simultaneous update. Have the units become different?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Initialisation and reproducibility

Reproduce two **training** traces by saving their distinct initial values.

The facts we will use

Supplied facts

Loss $J(a)=(a-2)^2$; gradient $2(a-2)$; rate 0.25; two updates. Restart from $a = 0$ and separately $a = 4$.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/17

Initialisation and reproducibility

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Start from 0

Updates: a = 1 then 1.5; final loss 0.25.

2. Start from 4

Updates: a = 3 then 2.5; final loss 0.25.

3. Record experiment

Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

CHANGE ONE THING

Repeat the a = 0 run with identical arithmetic and settings.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start from 0	Updates: a = 1 then 1.5; final loss 0.25.
2. Start from 4	Updates: a = 3 then 2.5; final loss 0.25.
3. Record experiment	Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

What this example does not establish

Keep the stated limit

A saved **seed** alone does not reproduce every software or hardware execution; record

environment and nondeterminism too.

Keep the stated limit

Read the labelled picture: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Repeat the a = 0 run with identical arithmetic and settings. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Initialisation and reproducibility claim and explain a repair using the supplied evidence.

Check the claim against the facts

For the two-unit example, y = a ReLU(ux) + b ReLU(vx). At x = 1, u = v = a = b = 1, **output** is 2 and target is 1. Because both ReLUs are in their positive region, changing any one parameter slightly changes y at rate 1.

Squared-error gradient is twice residual times that sensitivity, so each gradient is 2. Students can check it with the symmetric numerical probe from Week 10.

Check the claim against the facts

With identical initial weights, identical data and deterministic symmetric updates, these units remain copies and cannot learn distinct roles through those updates alone. Different hidden weights can break the **symmetry**.

For example, u = 1 and v = -1 give hidden activations [1, 0] at x = 1 and [0, 1] at x = -1. This demonstrates different responses, not a guarantee of successful **training**.

Check the claim against the facts

Zero is not always an invalid start: v = 0 worked in our single trained-**output**-weight model.

But setting all four weights to zero in the two-unit product model leaves every single-coordinate loss probe unchanged, because each path still contains a zero factor.

Do not generalise the success of one scalar start to every architecture.

Locate the earliest unsupported step

Both runs use the same **seed** and generator. Run A assigns the first four draws to its weights. Run B draws a preview number first, then assigns the next four. Must the initial weight arrays match? What should be saved?

State one repair and the checks you would rerun.

Practise together

> Practise Initialisation and reproducibility with support, then explain the deciding step in your own words.

For the scalar lesson model, start v = -2 and apply two updates at rate 0.1. Give all three v values and the final **training** MSE.

For x = 1, calculate the two-unit output and squared error before and after the shown simultaneous update. Have the units become different?

Explain your prediction to a partner before comparing results.

Practise with less help: Distinguish equality of loss and parameters

Use workbook W2 for the connected example "**Initialisation** and reproducibility". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Initialisation** and reproducibility. Proposed explanation: “The same **seed** specifies the entire experiment”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Initialisation and reproducibility. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

```
from random import Random

def initial_values(seed):
    rng = Random(seed)
    return [rng.uniform(-1.0, 1.0) for _ in range(4)]

def train(v):
    states = [v]
    for _ in range(2):
        v = v - 0.1 * 5 * (v - 2)
        states.append(v)
    return states

first = initial_values(17)
second = initial_values(17)
assert first == second
print(first)
for start in [0.0, 1.0, 3.0]:
    print(start, train(start))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Initialisation and reproducibility. Explain why each step is justified.

Explain → apply → challenge

For the scalar lesson model, start v = -2 and apply two updates at rate 0.1. Give all three v values and the final **training** MSE.

> Both runs use the same **seed** and generator. Run A assigns the first four draws to its weights. Run B draws a preview number first, then assigns the next four. Must the initial weight arrays match? What should be saved?

A forward-pass program returns the same wrong answer every run with a fixed **seed**. Has repeatability verified the implementation? Name two stronger checks.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **initialisation** and reproducibility **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how starting values affect a fixed **training** budget.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Save actual initial parameters, their order, data and split IDs, pre**process**ing rules, model formulas, loss convention, learning rate, batch order, update budget, selection policy, code version, software environment and final checkpoint.

A **seed** helps replay one source of randomness; it does not specify all of these choices or guarantee identical results across different numerical environments.

Two runs use seed 17 but have different final losses. One run made an extra random draw and used a different batch order. Is either difference evidence that the network is broken? Describe a useful investigation.

Repeatability alone cannot prove correctness: a program can repeat the same mistake. Preserve hand-calculated checks as well as logs.

When comparing starts, report every planned run instead of selecting a lucky result using final test outcomes. Next week locates shape and gradient errors.

A forward-pass program returns the same wrong answer every run with a fixed **seed**. Has repeatability verified the implementation? Name two stronger checks.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Initialisation and reproducibility case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Identical units receive identical updates

HIDDEN UNIT	BEFORE THE UPDATE	AFTER THE UPDATE
First ReLU unit	u is 1 and a is 1	u is 0.8 and a is 0.8
Second ReLU unit	v is 1 and b is 1	v is 0.8 and b is 0.8

In this separate two-unit example there are no biases. At x = 1, output is 2 and target is 1. Each of the four squared-error partial gradients equals 2. A simultaneous update at rate 0.1 changes every weight from 1 to 0.8 and keeps the units identical.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer the starting-value experiment independently

Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Initialisation and reproducibility, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Reproduce two **training** traces by saving their distinct initial values.

By the end, I can	How I will show it
Trace how starting values affect a fixed training budget	Explain the deciding step, show a trace or test, and state one limit.
Explain a simple hidden-unit symmetry	Explain the deciding step, show a trace or test, and state one limit.
Record enough information to replay and audit a classroom experiment	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Initialisation and reproducibility without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace how starting values affect a fixed **training** budget

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Reproduce two **training** traces by saving their distinct initial values.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

1. Record experiment

2. Start from 0

3. Start from 4

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record experiment Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Repeat the a = 0 run with identical arithmetic and settings. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. It again reaches 1 then 1.5 in this deterministic toy calculation.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It again reaches 1 then 1.5 in this deterministic toy calculation. Identical initial state and update rules determine the same arithmetic trace.

Repair a missing building block

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

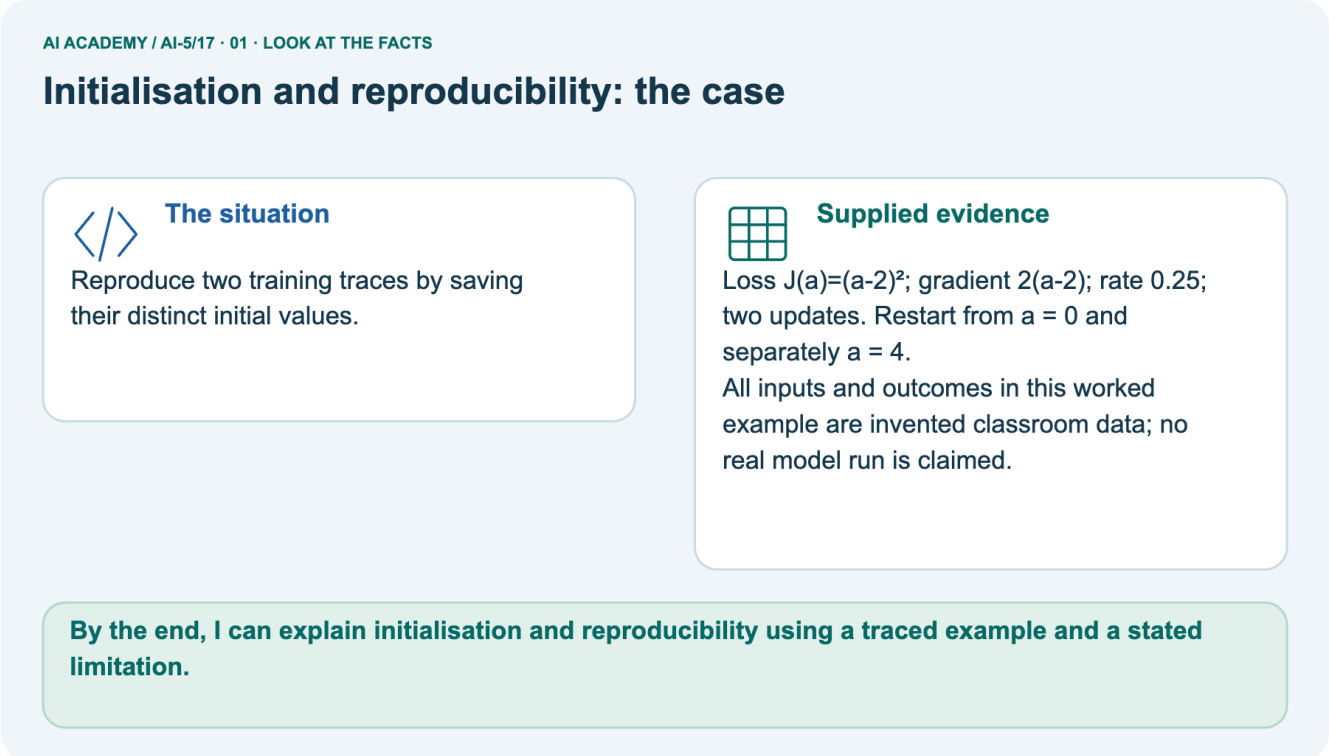
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Initialisation and reproducibility graphic using its labels, deciding evidence and relevant limitation.

01. Initialisation and reproducibility: the case



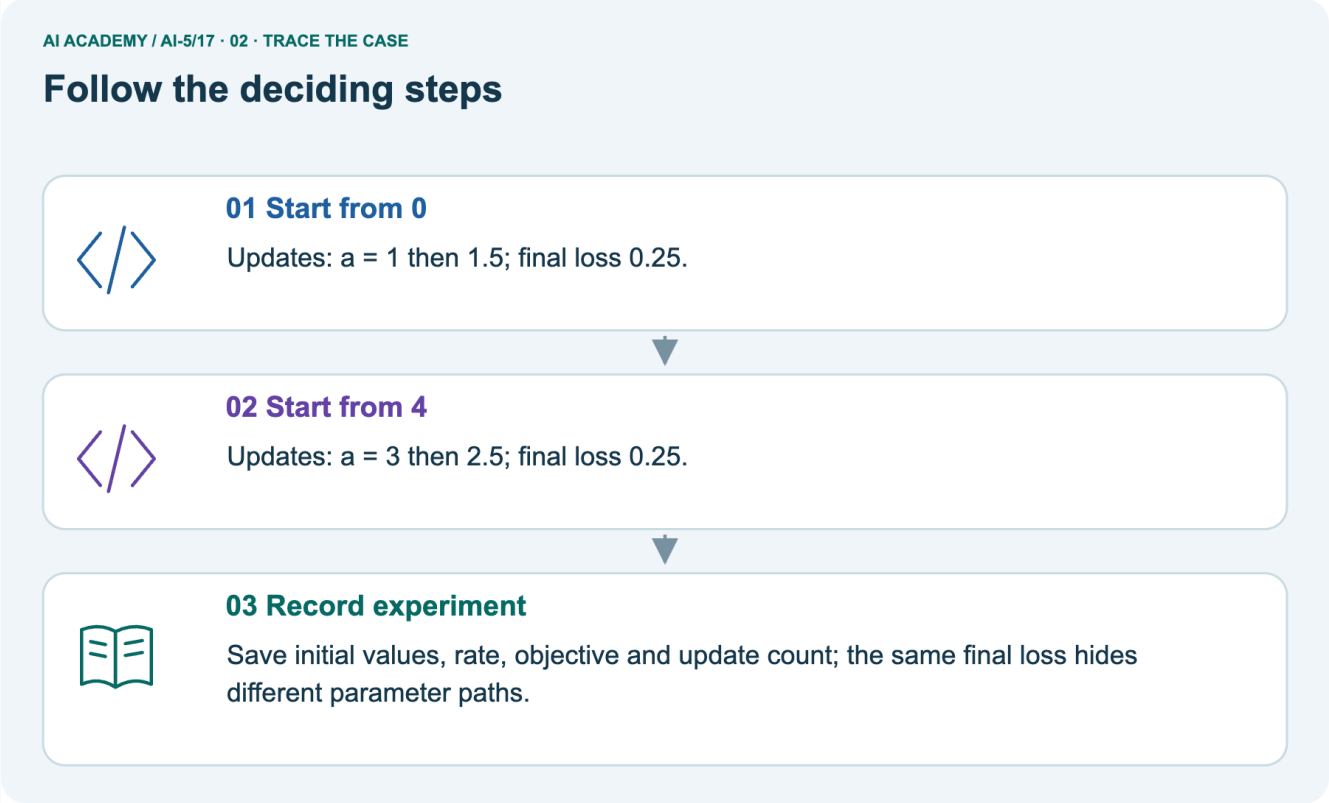
Follow the labelled picture with the matching explanation. Open the vector diagram

Reproduce two **training** traces by saving their distinct initial values. Supplied facts: Loss J(a)=(a-2)²; gradient 2(a-2); rate 0.25; two updates. Restart from a = 0 and separately a = 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

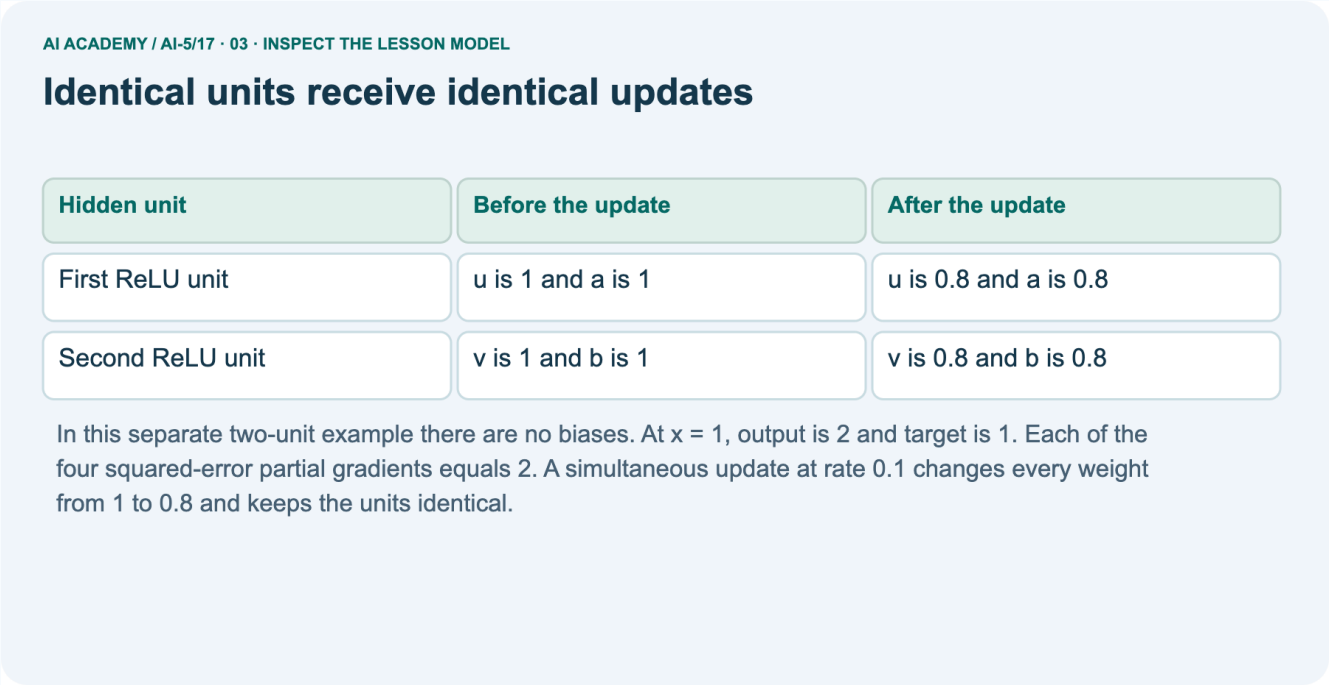


Follow the labelled picture with the matching explanation. Open the vector diagram

Start from 0: Updates: a = 1 then 1.5; final loss 0.25. → Start from 4: Updates: a = 3 then 2.5; final loss 0.25. → Record experiment: Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Point to each numbered stage and explain which recorded **input** it uses.

03. Identical units receive identical updates

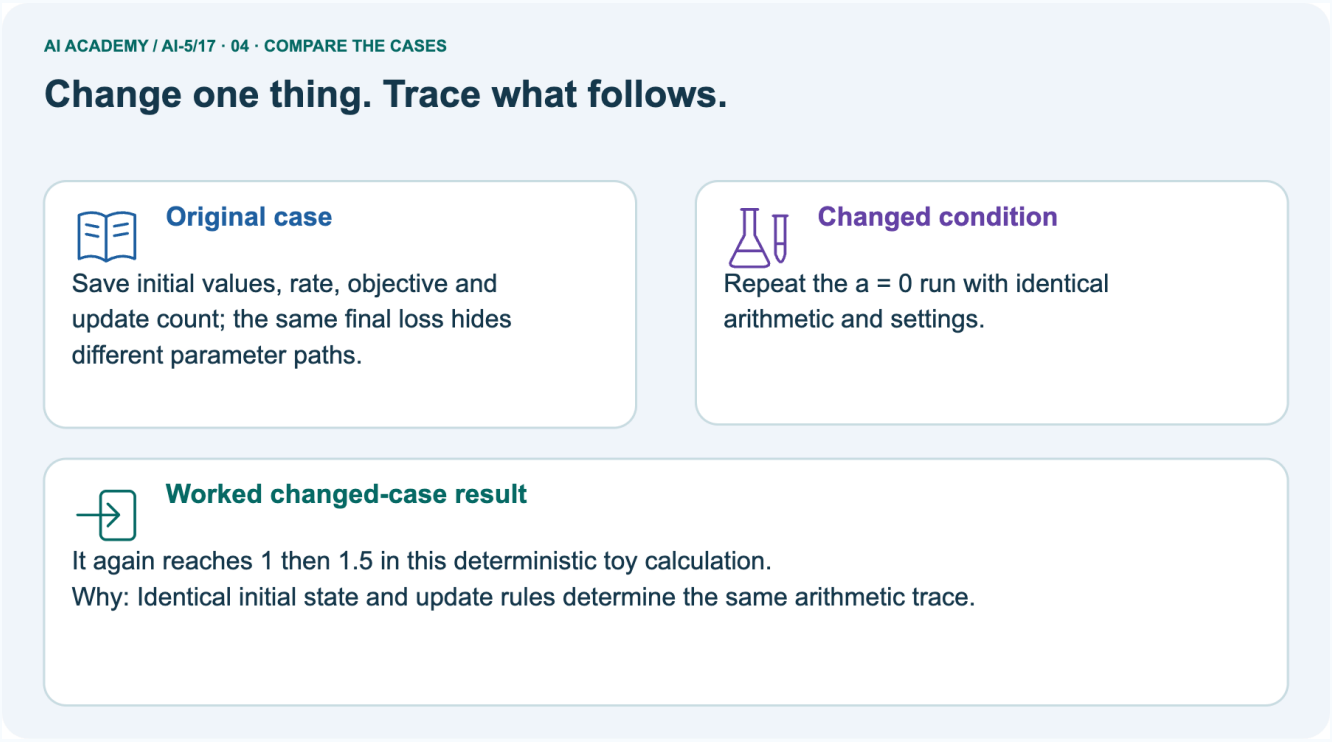


Follow the labelled picture with the matching explanation. Open the vector diagram

Identical units receive identical updates. First ReLU unit | u is 1 and a is 1 | u is 0.8 and a is 0.8. Second ReLU unit | v is 1 and b is 1 | v is 0.8 and b is 0.8

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



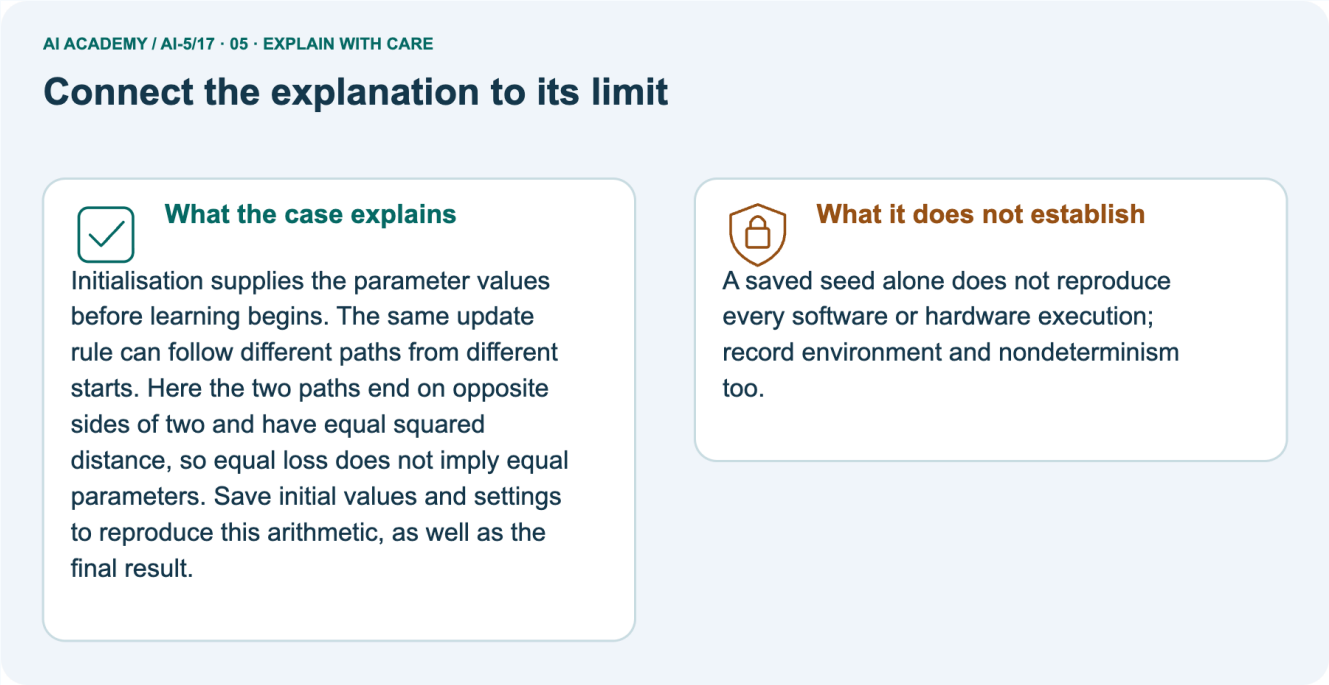
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Save initial values, rate, objective and update count; the same final loss hides different parameter paths. Changed condition: Repeat the a = 0 run with identical arithmetic and settings.

Expected result: It again reaches 1 then 1.5 in this deterministic toy calculation. Reason: Identical initial state and update rules determine the same arithmetic trace.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

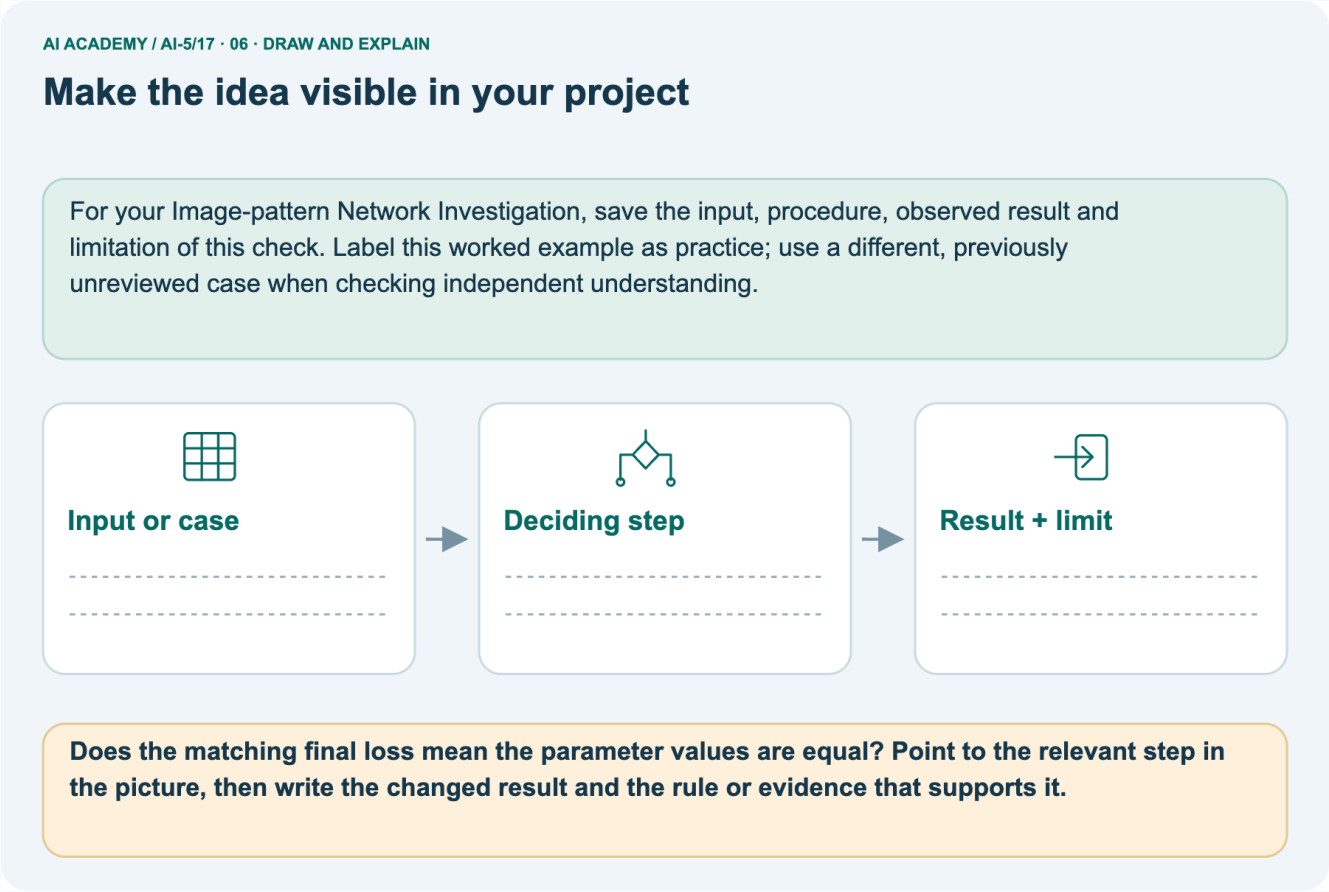
Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

Limit: A saved **seed** alone does not reproduce every software or hardware execution; record environment and nondeterminism too.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.